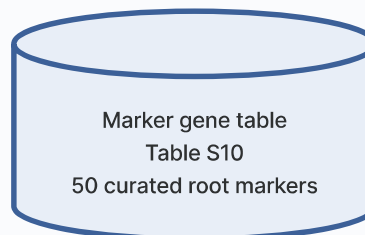
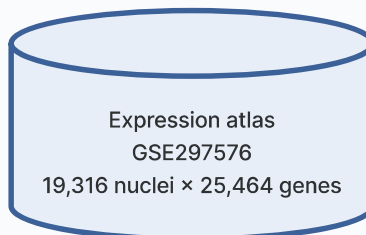
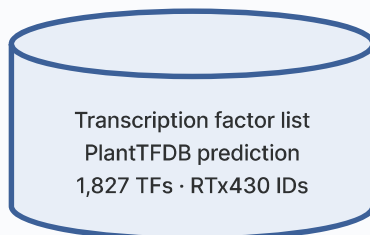


INPUTS · Subheading 2.5



USER · before the agent runs

Construct TF list in the
expression data's namespace
Subheading 3.2

INSPECT THE OBJECT
record expected values
assays · cell types · doublets · detection
thresholds **Subheading 3.3**

Write prompt.md
inputs · fixed parameters ·
expected counts · stop conditions
Subheading 3.4

Build and verify environment
Subheading 3.5 · synthetic test
returns 95 edges

prompt.md

AGENT · executes from prompt.md

Export and filter
Seurat object to h5ad, singlet nuclei
only **Subheading 3.6**

expect 14,792 singlet
nuclei

Prepare regulators present in
matrix and detected in ≥ 50
nuclei **Subheading 3.7**

expect 1,370 → 1,022
regulators

GRNBoost2 inference
per cell type · SLURM batch
jobs · seed 42 **Subheading 3.8**

3 jobs · ~1.4 M edges each

Rank regulators: importance summed over
cell-type marker genes **Subheading 3.9**

OUTPUTS · Subheading 3.10

Figures 2–5
UMAP · features
· dotplot ·
networks

Edge tables
3 × CSV

Ranked
regulators
top 10 per cell
type